Journal of the History of Biology (2008) 41:369–388
DOI 10.1007/s10739-007-9136-6

The Spatial Turn: Geographical Approaches in the History of Science

DIARMID A. FINNEGAN
School of Geography, Archaeology and Palaeoecology
Queen's University
Belfast, BT7 1NN
UK
E-mail: d.finnegan@Queens-Belfast.AC.UK

Abstract. Over the past decade or so a number of historians of science and historical geographers, alert to the situated nature of scientific knowledge production and reception and to the migratory patterns of science on the move, have called for more explicit treatment of the geographies of past scientific knowledge. Closely linked to work in the sociology of scientific knowledge and science studies and connected with a heightened interest in spatiality evident across the humanities and social sciences this 'spatial turn' has informed a wide-ranging body of work on the history of science. This discussion essay revisits some of the theoretical props supporting this turn to space and provides a number of worked examples from the history of the life sciences that demonstrate the different ways in which the spaces of science have been comprehended.

Keywords: historiography, science, sociology, spatiality

Glancing through the programs of conferences devoted to the history of science it is now commonplace to encounter sessions and papers organised around a concern with the spatiality of scientific knowledge. Place and space, territory and topography, cartography and scale, borders and boundaries are just some of the geographical terms commonly employed in titles which otherwise flag a more specific subject matter. The same kind of encounter could be replicated by a more strategic leafing through recent publications by historians of science. This essay offers a review of this 'geographical turn' by outlining some of the reasons for the infusion of spatial vocabulary into historical accounts of science. It then turns to work in the history of the life sciences that in a variety of ways has examined the historical geographies of scientific knowledge. As well as registering the diversity of approaches the worked examples will be used to point to some of the wider historiographical challenges raised by studies of the spaces of science.

Making Space for Space

The so-called spatial turn in the history of science has routinely been formulated alongside other theoretical commitments. Commonly, if not invariably, a concern with space in the study of science has been introduced as a corollary of a sociological analysis dealing with the indelible mark left by different social and cultural milieux on the making and marketing of scientific claims. The strong program's symmetry postulate – that any explanation of competing scientific claims should bracket a concern with ontological truth and ask the same sociological questions of local accounts of scientific credibility – has provided grounds to undermine automatic 'denigration by contextualisation' and to scrutinise rather than assume the disembodied or 'placeless' character of scientific rationality.[1] In this view, criteria for evidential warrant and experimental procedure are approached as local accomplishments rather than universal constants and precisely where scientific activities took place becomes an important subject for historical inquiry.

While the emphasis on science as a form of local knowledge has extended the explanatory reach of any geographical account of science it has also risked privileging the local and precluding larger-scale explorations. Recognising this, historians concerned with the sociology of science have argued that a focus on the local circumstances conditioning particular scientific projects, while useful in better understanding the socially-embedded character of scientific knowledge production and reception, is insufficient when attempting to understand the striking global success of science.[2] As they have pointed out, scientific knowledge apparently travels with unique efficiency and scientific practitioners harbour ambitions that patently extend far beyond local contexts. Beyond philosophical questions about the localist character of sociological studies conducted according to the symmetry principle, a concern with the circulation of scientific knowledge has encouraged historians to propose a number of approaches that encourage sociological and

[1] Bloor, 1997, p. 379; Ophir and Shapin, 1991, p. 4.
[2] Secord, 2004.

geographical accounts sensitive to the translocation of science.[3] Perhaps the most familiar is Steven Shapin's argument that the circulation of scientific knowledge is ineluctably bound to questions of trust.[4] It is not hard to see that the questions associated with work on trust and travel – how exactly 'distance lends enchantment', how it enhances or erodes credibility and shores up or subverts the epistemic and cultural authority of science – are profoundly geographical. It becomes necessary, as Jan Golinski has argued, to 'map the large-scale networks in which practitioners of the sciences are involved, [networks] which extend much further than those exploited by the members of other human cultures'.[5]

If the sociology of scientific knowledge has encouraged a close study of science's local operations and concomitant global ambitions it has also unsettled assumptions about a clear boundary between scientific work and other cultural domains. Accounts of the wider cultural consequences of scientific developments have opened up aspects of scientific culture beyond epistemic norms and cognitive practices. Such accounts of the 'active traffic' between scientific endeavour and other cultural projects have directed attention to a wider cultural geography of science not always apparent in studies focused on the social constitution of scientific credibility inside the walls of the laboratory or in studies concerned with the epistemic hurdles associated with transmitting science from place to place.[6] This is particularly evident in work that focuses on the reception of scientific ideas and the consumption of scientific texts.

The growing recognition of the need for studies sensitive to the historical geography of scientific knowledge has also opened up dialogue with sociologists, anthropologists and cultural geographers who previously treated science as a form of knowledge uniquely immune to fully-fledged socio-spatial analysis.[7] A central guiding assumption often sustaining and directing these conversations has been that space need not be thought of as a 'container' or 'backdrop' for social life – a view often described as an abstract 'Cartesian' notion of geometric space – but rather as an active ingredient in social and cultural life or an inescapable (which is not say uniform) mode of existence. This 'relational'

[3] On the symmetry principle from a (rather critical) philosophical perspective, see Lewens, 2005. For a recent defence of the strong program, see Bloor, 1997.

[4] For example, Ophir and Shapin, 1991, p. 15ff; Shapin, 1995, p. 307ff.

[5] Golinski, 2005, p. 172.

[6] On cultural studies of science, see Rouse, 1992.

[7] Evidence of these conversations can be found in general introductions to the 'spatial turn' found in a variety of disciplinary contexts. See, for example, Gieryn, 2000; Gunn, 2001; Yanni, 2005.

view of space has provided grounds for integrating more fully geographical and sociological accounts of science and has been widely adopted by scholars explicitly concerned with developing a historical geography of scientific knowledge.

This intellectual ferment has been lent further support by a number programmatic statements recommending more systematic attention to the spatiality of science.[8] None of these proposals claim to be exhaustive but each presents a provisional geographical framework or taxonomy of spatial categories offered as an aid for scholars concerned with the historical geographies of science. Common among such proposals is the suggestion that sites of scientific knowledge production – be they museums, gardens, laboratories, field stations or other less customary examples – provide a definite geographical focus for discerning *in situ* the closely linked spatial and social character of scientific practice without ignoring wider channels of scientific exchange. The acquisition of skill sets, the regulation of bodies and instruments and the face-to-face interaction of experts are among the social practices argued to be tightly bound up with the micro-geography of spaces cordoned off from an extra-scientific 'outside'.

The regional geography of scientific endeavour has also been tendered as a fruitful focus for work on the spaces of science enabling identification of features of scientific culture positioned between the local and the global. A regional focus, it has been advised, need not be restricted to a well-worn concern with national styles but can involve the study of the entanglements of science and regional identity.[9] This proposal allows the role scientific activities have played in the formation of regional identities to be examined and suggests that the regionality of science can be approached in a way that analyses rather than assumes the boundaries and character of different regional spaces.[10]

Territory is another category that has appeared in accounts recommending the advantages of a spatial history of scientific knowledge.[11] Unlike site and region, territory does not immediately imply a particular scale of inquiry and exploring the territorial aspects of science can direct attention to the more abstract spaces of disciplinary divisions of labour without losing sight of the material terrain necessary for securing,

[8] Harris, 1998; Livingstone, 1995, 2003; Naylor, 2005; Smith and Agar, 1998; Shapin 1998; Withers, 2002.

[9] For a sustained look at the regional geography of past science, see Livingstone, 2003, pp. 87–134.

[10] For two examples, see Raffles, 2001; Withers, 2001.

[11] See, in particular, Smith and Agar, 1998.

controlling and reinforcing epistemic and social demarcations.[12] Examining the territorial ambitions of science may also disclose diverse links between science and other enterprises with powerful vested interests in territorial expansion and control. Studying territories, whether intellectual, material or political, can also raise the issue of boundary construction supplying another geographical term that has been widely employed by scholars concerned with the manufacture and maintenance of scientific expertise and identity.[13]

Sites, regions, territories and boundaries imply, on first thought, a static account of the geography of science which may miss the geographies described by science on the move. When scientific knowledge travels it transmutes and huge investments of labour and resources are required to translate knowledge from one place to another in either literal or metaphorical terms.[14] For these and other reasons it has been suggested that the circulation of scientific knowledge, instruments, personnel and objects should be carefully charted and accounted for. Following this line of inquiry, Steven Harris has called for attention to the 'kinematic geography of movement' (the provenance and transport routes of the supporting materials necessary for science) and the 'dynamics of travel' (the modes and motivations of scientific travellers).[15]

Noticing site, region, territory and circulation is merely to skim off some of the more common categories commended by proponents of a spatialised historiography and other terms could be readily appended. Yet such an exercise must recognise the limitations of spatial categories defined independently of historical contingencies. As Charles Withers puts it:

> Asking the question 'What is meant by the historical geography of knowledge?' does not, initially, permit of a very precise answer since it will depend upon stating more clearly what exactly the knowledge in question is, at what scale of analysis it is to be understood and on the focus of inquiry – over time, at a 'moment', in given sites, of certain methods, in particular social networks and so on.[16]

The remaining part of the essay examines how an interest in the historical geography of science has been mobilised by historians of the life

[12] For accounts that utilise 'territory' in approaches the spaces of science, see for example Smith and Agar, 1998.

[13] For example, Davidovitch and Zalashik, 2006; Gieryn, 1999; Star and Griesemer, 1989.

[14] For a study of the literal translation of a scientific text, see Rupke, 2000.

[15] Harris, 1998, p. 273.

[16] Withers, 2001, p. 255.

sciences. In so doing I do not mean to suggest that accounts not giving explicit attention to space are necessarily insensitive to geography. To cite one classic example, David Allen's acclaimed social history of natural history could equally be described as a consummate social geography.[17] Rather, my purpose here is to highlight some of the ways in which historical geographers and historians of science have consciously grappled with and re-thought the spatiality of scientific knowledge. I begin with work that has examined the manufacture and modification of 'science *in situ*'. Here the focus is on formative places that influenced and resourced scientific work in different ways. While concentrating on particular sites and larger regions the assumption is not that science fashioned locally was local without remainder. Any study of science in its place rapidly reveals connections with other settings and wider concerns. In my second section, which explores the historical geographies of 'science in motion' the local does not disappear from view. The focus, however, is on accounts that deal with the spread of scientific knowledge, practices and objects and with the politics associated with the global ambitions of scientific endeavour.

Science *in situ*

Approached as a set of coordinated practical activities science can be described a situated and social enterprise. We might wish to press that claim further and argue that where those activities take place fundamentally shapes the direction and the content of scientific research. Understood as a historiographical principle this claim concentrates attention on the local character of any scientific enterprise. In this section I will focus on examples that to a greater or lesser extent are interpreted through this methodological lens.

Work on the historical geography of museums has supplied particularly telling examples of the relations between the concrete (sometimes literally) particularities of place and contingencies of scientific practice. Accounts of acrimonious struggles over museum space have demonstrated that the influence of built form on science was not incidental to disciplinary developments and the direction of scientific careers. The management of space at the Muséum National d'Histoire Naturelle under the direction of Georges Cuvier and the construction of Richard Owen's Natural History Museum in London are two thoroughly researched and closely connected examples of museum sites that

[17] Allen, 1976.

significantly altered the intellectual direction and social dynamics of 19th-century natural history.[18] Sophie Forgan's comparison of two London museums, Owen's Natural History Museum and the Museum of Practical Geology in Jermyn Street, lead her to conclude that the buildings 'were more than simply a shell within which scientists had to manage as best they might'.[19] The internal arrangements and architecture of the Natural History Museum and the Museum of Practical Geology provided a fixed and enduring structure for negotiating disciplinary territory. Whether the buildings and their internal spatial layouts were celebrated or castigated they provided an important material point of reference in the ongoing task of restructuring scientific research.

As well as shaping research practices and priorities museums and their buildings also helped create and control relations between science and wider society. The physical location of museum buildings often announced the significance of scientific knowledge for national life and local civic society. So, siting the building to house the natural history collections of the British Museum in South Kensington was part of a wider move to assemble London's scientific elite in a single salubrious urban space and involved a 'politics of location' connecting science with wider civic and national concerns.[20] The imposing architectural style of the Museum, while open to different readings, was put to good use by those concerned with the cultural authority of science.[21] The Museum building not only tempered the trajectories of disciplinary development but also enabled and regulated exchanges between science and civic society.

The regulation of mutually transforming exchanges between science and society in historical contexts can be further exemplified by reference to botanical gardens. It has been widely recognised that botanical gardens, functioning as imperial centres of accumulation and calculation and as important local cultural institutions, provided the necessary conditions for a global botanical science that carried a strong cultural and political charge.[22] Scientific gardens thrived not on insularity but on multiplying networks of exchange while preserving strategic connections

[18] On Owen's Natural History Museum, see Forgan, 1999 and Yanni, 1999; for Cuvier, see Outram, 1996 and Spary, 2000.

[19] Forgan, 1999, p. 203.

[20] On the politics of juxtaposition and separation evident in struggles over space in Victorian London, see Forgan, 1998.

[21] On this point, see Forgan, 2005, p. 576–577.

[22] For example, Brockway, 1979; Outram, 1996; Spary, 2000

with local cultural and political contexts. Overlapping concerns with cultural and scientific relevance were inscribed and reinforced more or less successfully in the space syntax of garden design. In the eighteenth and nineteenth centuries in particular botanical gardens provided an ordered space that reified conceptions of aesthetic and scientific propriety.[23] Gardens, like museums, were, in other words, polyvalent sites that tacked botanical science onto other sorts of political and cultural projects. The weight of symbolic meanings associated with the Jardin du Roi and its later incarnation as the Muséum National d'Histoire Naturelle provides one salient example of the culturally-charged character of scientific gardens. As Emma Spary has shown, the Jardins/ Muséum represented and reinforced at different points in its evolution royal rule, economic reform and republican polity. Such meanings and transformations were possible because the gardens provided a venue in which 'naturalists [could be] *bricoleurs* generating a wide variety of meanings and practices within a space that was very different from the homogeneous world of the laboratory'.[24]

Museums and botanic gardens, often proximate physically, scientifically and culturally, are only two of the more studied scientific sites that can be approached with geographical questions in mind. David Matless and Laura Cameron's cultural geography of an experimental pool dug in the Norfolk Broads in 1953 offers a more unconventional case study.[25] The pool, designed as a microcosm of the broads and created by the ecologist Marietta Pallis, carried other cultural meanings related to Pallis's interests in Bergsonian vitalism and Greek Orthodox Christianity. Cut to create among other symbolic forms a double-headed Byzantine eagle the pool confirmed Pallis's flamboyant scientific heterodoxy and her deliberate disavowal of a strict separation between ecological science and metaphysical contemplation. As an experimental field site the pool also became enrolled in scientific debate about the origins of the broads and was a cause and consequence of Pallis's own position within different scientific and cultural constituencies. For Matless and Cameron, the pool allowed Pallis to 'revision the regional landscape' and to negotiate in an idiosyncratic fashion the boundaries between science and culture.

Such high resolution accounts of science *in situ* can be supplemented by studies of the regional configuration of particular scientific projects.

[23] On scientific and aesthetic convention in garden design, see Johnson, 2006.

[24] Spary, 2000, pp. 246ff

[25] Matless and Cameron, 2006.

Alison Kraft and Samuel Alberti's account of British biology in the late-Victorian period provides one telling example of how regional culture can assume an important role in shaping a particular episode in the development of a scientific discipline.[26] With the aim of better understanding 'the broader institutional topography of British biology' Kraft and Alberti investigate the local and regional culture in which biological teaching and research in three civic colleges in the north of England took shape. Among other arguments, their paper suggests that a regional scientific culture that continued to value museum and field-based amateur natural history modulated and modified wider trends privileging the laboratory as a site of professional biological expertise. Laboratories co-existed with museums and the boundary between amateur and professional botanists remained blurred and porous in ways missed by accounts insensitive to local and regional differences.

The local and regional character of scientific endeavour is also apparent in studies of the reception of scientific texts. James Secord's meticulous reconstruction of the 'geographies of reading' evident in Victorian encounters with Robert Chambers's *Vestiges* uncovers a rich set of responses that varied not only across a social spectrum but also between the different urban and regional cultures associated with five British cities.[27] It becomes clear in Secord's account that reactions to Chambers's 'sensational' science were articulated in local and regional accents and were cross-stitched with local religious and cultural concerns. A similar concern with the geographies of reception is evident in Nicolaas Rupke's study of Alexander von Humboldt. In earlier work Rupke demonstrated through bibliometric analysis how von Humboldt's *Essai Politique* was differently reviewed in Germany, France and Britain.[28] More recently Rupke has extended his account of Humboldt by employing a 'metabiographical' approach that situates the different renditions of Humboldt's life in different cultural settings.[29] Innes Keighren's study of the reception of Ellen Churchill Semple's *Influences of Geographic Environment*, a seminal text implicated in the disciplinary development of early-20th-century Anglo-American academic geography, provides another cognate study that pays close attention to the geographies of reception.[30] For Keighren, in the act of reading meanings are made rather than passively received and are cross-referenced

[26] Kraft and Alberti, 2003.

[27] Secord, 2000, pp. 155–335.

[28] Rupke, 1999.

[29] Rupke, 2006.

[30] Keighren, 2006, p. 537.

with other texts and debates in ways that can be fruitfully analyzed with geographical questions in mind.

Accounts of the geographies of science printed and read might be usefully supplemented by narratives of geographies of science spoken and heard. This point has been most effectively made by David Livingstone in work on the regional geography of responses to Darwin. The varied reception of Darwinism in Belfast, Edinburgh and Princeton provides Livingstone with a vivid illustration of the geographies of scientific speech.[31] In 1874, John Tyndall's Presidential address to the British Association, meeting that year in Belfast, produced what Livingstone terms 'a distinctive rhetorical space' in which arguing for the compatibility of Presbyterian belief and evolutionary science was 'unsayable' if not unthinkable.[32] Tyndall's lecture sounded a materialist note, at least in the ears of Belfast Presbyterians, and rendered theologically suspect any local attempt to work out a more conciliatory approach to Christian religion and Darwinian science. In Princeton and Edinburgh – and in other Irish contexts like Derry and Dublin – Presbyterian convictions and Darwinism could, in contrast, co-exist publicly on more friendly terms. Even so, the reasons for this conciliatory atmosphere varied from city to city and region to region and this diversity was as apparent in how Darwinism was announced and heard as it was in how Darwinism was read and reviewed.[33]

Science in Motion

So far the primary assumption has been that science is formed *in situ* or that scientific knowledge 'possesses its shape ... by virtue of the physical, social, and cultural circumstances in which it is made, and in which it is used'.[34] Yet science, as the foregoing examples already indicate, also travels from place to place in prefabricated and more malleable forms. Laboratory science, regulated by a concern with knowledge claims that can be reproduced wherever the same experimental conditions and procedures are faithfully re-enacted, depends upon the rapid and secure

[31] Livingstone, 1992; 2007.

[32] Livingstone, 2007.

[33] For extensions of this work to other settings (in Russia, New Zealand and the American South), see Livingstone, 2006.

[34] Ophir and Shapin, 1991, p. 4.

transfer of information, specimens, standards and expertise. Other scientific activities work with objects that cannot be readily controlled or codified within fixed spaces or channeled along trusted routes. Gregg Mitman's account of efforts to understand, monitor and treat human allergies provides a dramatic illustration of the way the objects of scientific inquiry can cut across more static or carefully regulated spaces of science. Pollen, Mitman reminds us, 'travelled within and across regional and disciplinary boundaries, ignoring the distinctions drawn by historians of 20th-century life science between biomedicine and natural history, the laboratory and the field'.[35] In light of this, Mitman argues for an 'aerial view of ecology' rather than one that 'peers out of the keyhole of laboratory life'.[36] It would seem that, in certain cases at least, we would be well advised to heed G. H. T. Kimble's critique of early-20th-century regional geography and attend to the links as much as the breaks in scientific landscapes.[37]

Mitman's advice is particularly pertinent for work on the historical geography of field science. In the field bodies as well as scientific objects are in constant motion and fieldworkers find themselves '*mobilis in mobile*', changing within change.[38] Further, field sites themselves are often transient and unstable spaces which emerge in the course of scientific investigation. This understanding of fieldwork has a long history and has been used to defend as well as denigrate the value of encountering nature outside controlled and artificial indoor spaces. Dorinda Outram, in a much-cited piece on the 'spaces of natural history', has drawn attention to George Cuvier's use of this image to defend his sedentary scientific habits. The *naturaliste-voyageur*, Cuvier noted, 'passes through, at greater or lesser speed, a great number of different areas, and is struck, one after the other, by a great number of interesting objects and living things.'[39] According to Cuvier, to 'roam freely about the universe' it was necessary to remain 'in one's study' and pursue truth systematically and unflinchingly. The sharp lines drawn by Cuvier between the study and the field were of course blurred in practice. As noted above, the institutional spaces in which Cuvier worked were in fact fluid and composite rather than stable and univocal. Even so, the epistemic and cultural challenges of fieldwork are such that Cuvier's distinction need not be dissolved entirely.

[35] Mitman, 2003, p. 506.
[36] Mitman, 2003, p. 507.
[37] Kimble, 1951.
[38] Verne, 1998, p. 47; p. 392–393, n. 47. On fieldwork, see Kuklick and Kohler, 1996.
[39] Cited in Outram, 1996, p. 259.

Relying on descriptions of fieldwork in a later period, Robert Kohler has proposed the shifting 'border' between fieldwork and laboratory work as 'one of the most important in the cultural geography of modern science'.[40] Painting in broad brush for heuristic purposes Kohler suggests that fieldwork is concerned with the messy particularities of different places and laboratories are, in contrast, 'placeless places'. It follows that concepts developed in social studies of laboratory science tend to be 'too abstract and timeless' to capture the contingent and dynamic character of field science.[41] Taking Kohler's lead, field sites might be approached not as fixed points in space but as 'working objects' differently constituted in different spaces of inquiry and fashioned by 'practices of place' such as pinpointing, demarcating, describing and sampling.[42]

For Kohler the borderland between laboratory and field provides a particularly important conceptual focus for understanding the changing character of North American biology before the Second World War. Focusing in particular on the 'border biology' of ecologists, Kohler tracks the dialectic between the placeless ambitions of laboratory practice and the exigencies of field practice. Occupying a kind of unstable hinterland between the laboratory and the field, border biologists 'experienced lab and field as movement and border crossing', a dynamic that generated impermanent hybrid spaces such as biological field stations and vivaria.[43]

While concentrating his analytical energies on the 'intangible cultural space' located between lab and field Kohler also underlines the importance of factoring in the more concrete places encountered and surveyed by the fieldworker. This nod towards 'physicality' has been repeated by others and has directed attention to a fruitful overlap with environmental history. As one group of historians have recently suggested:

> Environmental history through its penchant for broad historical narratives that give agency to both nature and humans, offers a compelling yet underutilised model to historians of science interested in connected their field's preoccupation with local sites of knowledge production to narratives that reach across larger spatial and temporal scales.[44]

[40] Kohler, 2002, p. 1.

[41] Kohler, 2002, p. 14.

[42] On 'working objects' in science, see Daston and Galison, 1992, p. 85. For work that emphasises the 'distributed' and polycentric nature of field sites, see Kennedy, forthcoming and Lorimer and Spedding, 2005. The phrase 'practices of place' is Kohler's.

[43] Kohler, 2002, p. 12.

[44] Mitman et al, 2004, p. 4

One example of work in this vein is Matthew Evenden's environmental history of the Fraser River in British Columbia, Canada.[45] Among other things Evenden argues that fisheries science in British Columbia was shaped by, as well as deeply involved with, the environmental history of the river. Key actors in debates about the potential effects of damming the river biologists, partly in response, developed an applied fishery science concerned with the life cycles and migratory patterns of salmon. To pursue this end involved the setting up of large-scale field experiments and the tracking of salmon over long distances. As well as 'developing ideas *in situ* in response to local conditions', the fishery scientists had to cope with the problems investigating the movement of salmon in a location peripheral to centres of scientific expertise.[46] In Thomas Gieryn's terms, to make the Fraser River a recognised 'truth spot' it was necessary that its peculiar character (its environmental history) and its representativeness were given due attention.[47]

In striving to overcome as well as exploit locality in order to capture wider movements or manufacture general claims science has frequently fed off and reinforced wider geopolitical ambitions. Studies of the relations between biogeography and empire in the nineteenth century have provided some intriguing accounts of the geographies of scientific activity working at a global scale. Janet Browne, for example, has sketched the links between plant and animal geography and the spatial economy of empire during the eighteenth and nineteenth centuries.[48] Travelling naturalists, concerned with the global distribution of flora and fauna, relied upon the infrastructure of colonial expansion and borrowed the language of imperialism to describe plant and animal geography. As Emma Spary puts it, many globe-trotting naturalists were embroiled in the conjoined 'micropolitcs of botanical identity and macropolitics of 18th-century colonial and social life'.[49]

In a finely argued piece, James Moore has developed such thoughts by demonstrating how the terminology of biogeography refracted and reinforced the diverse estimations of empire evident in the political and cultural constituencies of the naturalists' home countries. In particular Moore shows how different appraisals of imperialism were evident in

[45] Evenden, 2004a.
[46] Evenden, 2004b, p. 370.
[47] Gieryn, 2006.
[48] Browne, 1996.
[49] Spary, 2005, p. 188.

the evolutionary biogeography of Charles Darwin and Alfred Russel Wallace.[50] Darwin's liberalism occasioned an account of divergent evolution laced with imperial overtones which, on the surface at least, was consonant with the progressivist logic of British colonialism. Wallace, occupying a different political station and encountering nature 'in native boats and huts rather than aboard a naval hydrographic vessel', made space for a critique of rampant colonialism through an alternative account of plant and animal geography that muffled the militaristic march of life triumphant on full display in Darwin's biogeography.[51]

Studies of biogeography and empire have also suggested that colonial science cannot be understood as the local enactment of fixed scientific procedures and vocabulary invented and ratified in a metropolitan centre. Recent work on what Lorna Schiebinger calls 'biocontact zones' has pointed to the polycentric and hybrid character of colonial science.[52] Drawing attention to the fragility of natural historical investigations conducted on the fringes of imperial spheres of influence these accounts have unfolded the transcultural spaces that shaped the identity and science of travelling botanists. Keeping a keen eye on the material culture of colonial botany including the trading, exploitation, transporting and processing of specimens, the 'pidgin' character of colonial botany has been emphasised highlighting the influence of non-European knowledge systems on so-called metropolitan science.

Similar concerns with the connections between global ambitions, local negotiations and science are apparent in work on more recent developments in the life sciences. Bronwyn Parry, for example, has examined the geography of new biological products associated with the burgeoning life sciences industry and has charted the spaces of the accompanying renascence in plant collecting.[53] Investigating what Parry terms 'new centres of calculation' represented by the laboratories of large pharmaceutical corporations her work sheds light on the economic and moral geographies of the global trade in bio-information. As Parry points out, issues of ownership and reward make the precise movements of biological materials from source to consumer a crucial concern. Monitoring the transmission and multiple uses of bio-informational products has been beset with difficulties and has had knock-on effects in terms of regulating 'compensatory flows' towards donor countries in the

[50] Moore, 2005

[51] Moore, 2005, p. 121.

[52] Schiebinger, 2005, pp. 125ff.

[53] Parry, 2006.

form of royalties. This in turn has presented significant challenges to the geographical frameworks underpinning the standard principles of distributive justice. Parry's account thus keenly demonstrates how geographical considerations have played a crucial role in the development and regulation late-20th-century biology.

One final example of the geopolitics of science in motion can be found in Nicholas King's account of the 'emerging diseases' paradigm among North American virologists and public health officials in the 1990s.[54] King pays particular attention to what he terms the 'scalar narratives' that underpinned attempts to secure public support for particular biomedical research programs. Switching dramatically from global pandemics and serious risks to the health of the American nation to the micro-scale of new pathogens, advocates of the 'emerging diseases' campaign 'bypassed the messiness of specific locations' and shored up the cultural authority of the suite of actors and institutions with a stake in the public health industry. King's approach thus uncovers the politics of science in motion and the role of spatial rhetoric in reinforcing the cultural authority of health science.[55]

Conclusion

A guiding assumption that threads its way through the work summarised above is that science depends on the manufacture and management of different spaces – real or imagined – to accomplish its objectives and establish its credentials. The individual case studies have also thrown up a variety ways in which the spaces of science can be comprehended. On one account the spaces of science are tangible places that condition cognitive content and sanction scientific authority. On another they are social or cultural spaces that both constrain and enable particular kinds of scientific practices. On yet another account spaces are rhetorical or imaginative constructs used to ratify the public credibility of scientific institutions.

Reflecting on this diversity, two issues present themselves as critical for historical accounts sensitive to the geographies of scientific knowledge. The first is the question of agency. This question can be posed in different ways pointing to some of the choices to be made when employing spatial vocabulary. For example, should an accent be placed on the conditioning effects of different spaces of science or should the geographical agency of science itself be placed in the foreground? It is

[54] King, 2004.

[55] A similar tactic is employed by Kirsch, 2004.

not, of course, possible to resolve this and in one sense it is a false opposition but the question directs attention to the different interpretative strategies that can be employed in constructing historical geographies of science. It may be important to try and isolate those spaces that shape the kind of scientific claims or conduct under scrutiny. In other cases the spaces of science will be much more part of the action, integral to the very issues under dispute.[56]

Another rather different question about the relative importance of different kinds of spaces might be posed. For some historians, material objects and physical environments may prove to be crucial in understanding a particular scientific project. Here, material agency will be a key concern and questions may be raised about the fit between a geographical account of science and the sociology of scientific knowledge.[57] For others the spaces will be socially constituted and material agency will appear as a social construct produced by the actors under investigation. If there is a trend in recent literature it is towards taking more seriously the cultural geography of 'things' and focusing on the material culture of scientific praxis, a trend that has consequences for the kinds of spatial questions asked of any scientific enterprise.[58]

The second and related issue basic to any geographical account of science is that of scale. As with particular spaces, scale can be regarded as an independent or dependent variable. It can also feature, as King's account demonstrates, in the spatial representations used to legitimise particular scientific projects. Here again there are choices to made. Such decisions cannot be prescribed and it is important to recognise the advantages and disadvantages of following a particular line. To give one example, discounting the meso-scale and concentrating on the movement between the local and the global may miss the ways in which science becomes entangled with national concerns or regional identities. It is nevertheless important to recognise that different scales of analysis cannot be presumed to be distinct and stable. Rather, they may be better thought of as intertwined in ways that require close investigation. Scaling up from a local or regional account of science to the transnational and global need not entail a move away from the local or regional. 'Big science' might be approached not as a single and monolithic entity uniformly stretched across global space but rather as a dynamic conglomeration of practices, materials and people differently assembled in different places and relying on the translation and

[56] See Matless 2003.

[57] See, for example, Keeling 2007.

[58] Alberti 2005; Dritsas 2005.

transformation – more than straightforward diffusion – of data and theories. The implications of this are that the local and regional are not fixed points or bounded territories but rather instantiations of wider networks and flows.

The turn to space in the history of science, and in the humanities and social sciences more generally, is now some distance behind us. The reception of this theoretical innovation among historians of science has its own geography and it has not been possible here to comprehensively chart the successive modifications and variations that have been composed out of earlier suggestions that space matters in the making and mobilisation of scientific knowledge. The longevity of a concern with the spaces of science among historians of science and the proliferation of studies that in some way concern themselves with the spatiality of scientific knowledge nevertheless suggests that geographical motifs will continue to guide and inspire new work on the historical sociology of science.

References

Alberti, Samuel J. M. M. 2005. "Objects and the Museum." *Isis* 96: 559–571.

Allen, D. E. 1976. *The Naturalist in Britain*. London: Allen Lane.

Bloor, David. 1997. "Remember the Strong Program?" *Science, Technology and Human Values* 22: 373–385.

—— 2007. "Ideals and Monisms: Recent criticisms of the Strong Programme in the Sociology of Knowledge." *Studies in History and Philosophy of Science* 38: 210–234.

Brockway, Lucile. 1979. *Science and Colonial Expansion: The Role of the British Botanic Gardens*. New York and London: Academic Press.

Browne, Janet. 1996. "Biogeography and Empire." Nicholas Jardine, James A. Secord and Emma C. Spary (eds.), *Cultures of Natural History*, Cambridge: Cambridge University Press, pp. 305–321.

Davidovitch, Nadav and Zalashik, Rakefet. 2006. "Medical borders: historical, political and cultural analyses." *Science in Context* 19: 309–316.

Daston, Lorraine and Galison, Peter. 1992. "The Image of Objectivity." *Representations* 40: 81–128.

Dritsas, Lawrence. 2005. "From Lake Nyassa to Philadelphia: A Geography of the Zambesi Expedition, 1858–64." *British Journal for the History of Science* 38: 35–52.

Evenden, Matthew D. 2004a. "Locating Science, Locating Salmon: Institutions, Linkages and Spatial Practices in Early British Columbia Fisheries Science."*Environment and Planning D: Society and Space* 22: 355–372.

—— 2004b. *Fish Versus Power: An Environmental History of the Fraser River*. Cambridge: Cambridge University Press.

Forgan, Sophie. 1998. "But indifferently lodged'…: Perception and Place in Building for Science in Victorian London." C. Smith and J. Agar (eds.), *Making Space for Science: Territorial Themes in the Shaping of Knowledge*. Basingstoke: Macmillan, pp. 195–215.

—— 1999. "Bricks and Bones: Architecture and Science in Victorian Britain." P. Galison and E. Thomspon (eds.), *The Architecture of Science*. Cambridge, Massachusetts: MIT Press, pp. 181–208.
—— 2005. "Building the Museum: Knowledge, Conflict and the Power of Place." *Isis* 96: 572–585.
Gieryn, Thomas F. 2000. "A Space for Place in Sociology." *Annual Review of Sociology* 26: 463–496.
—— 1999. *Cultural Boundaries of Science: Credibility on the Line*. Chicago: University of Chicago Press.
—— 2006. "The City as Truth Spot: Laboratories and Field Sites in Urban Studies." *Social Studies of Science* 36: 5–38.
Golinski, Jan. 2005. *Making Natural Knowledge: Constructivism and the History of Science*, 2nd ed. Chicago: University of Chicago Press.
Gunn, Simon. 2001. "The Spatial Turn: Changing Histories of Space and Place." S. Gunn and R. J. Morris (eds.), *Identities in Space: Contested Terrains in the Western City since 1850*. Aldershot: Ashgate, pp. 1–14.
Harris, Steven. 1998. "Long Distance Corporations, Big Sciences, and the Geography of Knowledge." *Configurations* 6: 269–305.
Johnson, Nuala J. 2006. "Cultivating Science and Planting Beauty: The Spaces of Display in Cambridge's Botanic Gardens." *Interdisciplinary Science Reviews* 31: 42–57.
Keeling, Arn M. 2007. "Charting Marine Pollution Science: Oceanography on Canada's Pacific coast, 1938–1970." *Journal of Historical Geography* 33: 403–428.
Keighren, Innes M. 2006. "Bringing Geography to the Book: Charting the Reception Of Influences of Geographic Environment." *Transactions of the Institute of British Geographics* 31: 525–540.
Kennedy, Alasdair. Forthcoming. "In search of True Prospect: Making and Knowing the Giant's Causeway as a Field Site in the Seventeenth Century." *British Journal for the History of Science*.
Kimble, George H. T. 1951. "The Inadequacy of the Regional Concept." D. L. Stamp and S. W. Wooldridge (eds.), *London Essays in Geography: Rowell Jones Memorial Volume*. London: Longmans, pp. 151–174.
King, Nicholas B. 2004. "The Scale Politics of Emerging Diseases." *Osiris* 19: 62–76.
Kirsch, Scott. 2004. "Harold Knapp and the Geography of Normal Controversy: Radioiodine in the Historical Environment." *Osiris* 19: 167–181.
Kohler, Robert E. 2002. *Landscapes and Labscapes: Exploring the Lab-Field Border in Biology*. Chicago: University of Chicago Press.
Kraft, Alison and Alberti, Samuel J. M. M. 2003. "'Equal Though Different': Laboratories, Museums and the Institutional Development Of Biology In Late-Victorian Northern England." *Studies in the History and Philosophy of Biological and Biomedical Sciences* 34: 203–236.
Kuklick, Henrika and Kohler, Robert E. 1996. "Science in the Field: Introduction." *Osiris* 11: 1–14.
Lewens, Tim. 2005. "Realism and the Strong Program." *British Journal for the Philosophy of Science* 56: 559–577.
Livingstone, David N. 1992. "Darwinism and Calvinism: the Belfast–Princeton connection." *Isis* 83: 408–428.
—— 1995. "The Spaces Of Knowledge: Contributions Towards a Historical Geography of Science." *Environment and Planning D: Society and Space* 13: 5–34.

—— 2003. *Putting Science in its Place: Geographies of Scientific Knowledge*. Chicago: University of Chicago Press.
—— 2006. "The Geography of Darwinism." *Interdisciplinary Science Reviews* 31: 32–41.
—— 2007. "Science, Site and Speech: Scientific Knowledge and the Spaces of Rhetoric." *History of the Human Sciences* 20: 71–98.
Matless, David and Cameron, Laura. 2006. "Experiment in Landscape: the Norfolk Excavations of Marietta Pallis." *Journal of Historical Geography* 32: 96–126.
Matless, David. 2003. "Original Theories: Science and the Currency of the local." *Cultural Geographies* 10: 354–378.
Mitman, Gregg. 2003. "Natural History and the Clinic: The Regional Ecology of Allergy in America." *Studies in History and Philosophy of Biological and Biomedical Sciences* 34: 491–510.
Mitman, Gregg, Murphy, Michelle and Sellers, Christopher. 2004. "Introduction: A Cloud Over History." *Osiris* 19: 1–17.
Moore, James. 2005. "Revolution of the Space Invaders: Darwin and Wallace on the Geography of Life." D. N. Livingstone and C. W. J. Withers (eds.), *Geography and Revolution*. Chicago: University of Chicago Press, pp. 106–132.
Naylor, Simon. 2005. "Introduction: Historical Geographies of Science – Places, Contexts, Cartographies." *British Journal for the History of Science* 38: 1–12.
Ophir, Adi and Shapin, Steven. 1991. "The Place of Knowledge: A Methodological Survey." *Science in Context* 4: 3–21.
Outram, Dorinda. 1996. "New Spaces in Natural History." Nicholas Jardine, James A. Secord and Emma C. Spary (eds.), *Cultures of Natural History*, Cambridge: Cambridge University Press. pp. 249–65.
Parry, Bronwyn. 2006. "New spaces of biological commodification: the dynamics of trade in genetic resources and 'bioinformation'." *Interdisciplinary Science Reviews* 31: 19–31.
Raffles, Hugh. 2001. "The use of butterflies." *American Ethnologist* 28: 513–548.
Rouse, Joseph. 1992. "What are cultural studies of scientific knowledge?" *Configurations* 1: 57–94.
Rupke, Nicolaas. 1999. "A Geography of Enlightenment: The Critical Reception of Alexander von Humboldt's Mexico work." D. N. Livingstone and C. W. J. Withers (eds.), *Geography and Enlightenment*. Chicago: University of Chicago Press, pp. 319–340.
—— 2000. "Translation Studies in the History of Science: The Example of *Vestiges*." *British Journal for the History of Science* 33: 209–22.
—— 2006. *Alexander von Humboldt: A Meta-Biography*. New York: Peter Lang.
Schiebinger, Londa. 2005. "Prospecting for Drugs." L. Schiebinger and C. Swan (eds.), *Colonial Botany: Science, Commerce and Politics in the Early Modern World*. Philadelphia: University of Pennsylvania Press, pp. 119–133.
Secord, James A. 2000. *Victorian Sensation: The Extraordinary Publication, Reception, and Secret Authorship of* Vestiges of the Natural History of Creation. Chicago: University of Chicago Press.
—— 2004. "Knowledge in Transit." *Isis* 95: 654–672.
Shapin, Steven. 1995. "Here and Everywhere: Sociology of Scientific Knowledge." *Annual Review of Sociology* 21: 289–321.
—— 1998. "Placing the View from Nowhere: Historical and Sociological Problems in the Location of Science." *Transactions of the Institute of British Geographers* 23: 5–12.
Smith, Crosbie and Agar, Jon. (eds.) 1998. *Making Space for Science: Territorial*

Themes in the Shaping of Knowledge. London: Macmillan.
Spary, Emma. 2000. *Utopia's Garden: French Natural History from Old Regime to Revolution*. Chicago: University of Chicago Press.
—— 2005. "Of Nutmegs and Botanists." L. Schiebinger and C. Swan (eds.), *Colonial Botany: Science, Commerce and Politics in the Early Modern World*. Philadelphia: University of Pennsylvania Press, pp. 187–203.
Star, Susan L. and Griesemer, James R. 1989. "Institutional Ecology, 'Translations' and Boundary Objects: Amateurs and Professionals in Berkeley's museum of vertebrate zoology, 1907–39." *Social Studies of Science* 19: 387–420.
Verne, Jules. 1998 [1871]. *Twenty Thousand Leagues Under the Seas*. Oxford: Oxford University Press.
Withers, Charles W. J. 2001. *Geography, Science and National Identity: Scotland since 1520*. Cambridge: Cambridge University Press.
—— 2002. "The Geography of Scientific Knowledge." Nicolaas Rupke (ed.), *Göttingen and the Development of the Natural Sciences*. Göttingen: Wallstein, pp. 9–18.
Yanni, Carla. 1999. *Nature's Museums: Victorian Science and the Architecture of Display*. Baltimore, MD: Johns Hopkins University Press.
—— 2005. "History and sociology of science: interrogating the spaces of knowledge." *Journal of the Society of Architectural Historians* 64: 423–425.